Programming Fundamental lab "2"

Task 1:

Type and save the following programs in visual studio. Run these programs and observe their output.

Input I

```
#include<iostream>
 using namespace std;
 int main()
{
 char letter;
 letter = 'A';
 cout<<letter<<'\n';
  letter= 'B';
 cout<<letter<<'\n';
 cin>>letter;
 return 0;
}
```

Output I

```
A
B
```

Input II

```
#include<iostream>
 using namespace std;
 int main()
{
 char letter;
  letter = 65;
 cout<<letter<<'\n';
 letter = 66;
 cout<<letter<<'\n';
 cin>>letter;
  return0;
}
```

Output II

```
A
B
```

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Input III

```
#include<iostream>
 Using namespace std;
int main()
{
 int letter;
 letter = 'A';
 cout << letter << '\n';
 letter = 'B';
 cout << letter << '\n';
```

Output III

```
65
66
```

Task 2:

Take two integers as input from the user apply arithematic operations on them (+,-,*,/) as print them on screen.

Input:

```
#include<iostream>
 Using namespace std;
  int main()
  {
   int a,b;
   cout<< "Enter Fost number:";
   cin>>a; //Example : 10
   cout <<"Enter second number: ";
   cin>>b; // Example: 5
   cout<<"Sum="<<a+b<<endl; // 10+5=15
   cout<< "Substraction="<<a-b<<endl; //10-5=5
   cout << "Multiplication="<<a*b<<endl; // 10*5=50
   if (b!=0)
      cout<<"Division="<<a/b<<endl; // 10/5=2
      else
        cout<<"Division=Not possible(divided by zero)"<<endl;
       return 0;
  }
```

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Home task 2

Small to capital. (ASCII)

Input:

```
#include <iostream>
Using namespace std;
int main ( ) {
    char small;
cout << " Enter small letter: ";
        cin >> small;
char capital = small - 32; // ASCII trick:
  a = 97, A = 65, dif 32
cout << "Capital:" << capital;
        return 0;
}
```

Date:

Coding C++

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RUN MENU

```
{
    char letter;

    // Part 1
    letter = 'A';
    cout << letter << endl;

    letter = 'B';
    cout << letter << endl;

    // Part 2
    letter = 65;
    cout << letter << endl;

    letter = 66;
    cout << letter << endl;

    // Part 3
    int number;

    number = 'A';
    cout << number << endl;

    number = 'B';
    cout << number << endl;

    return 0;
}
```

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Compile Result

```
A
B
A
B
65
66

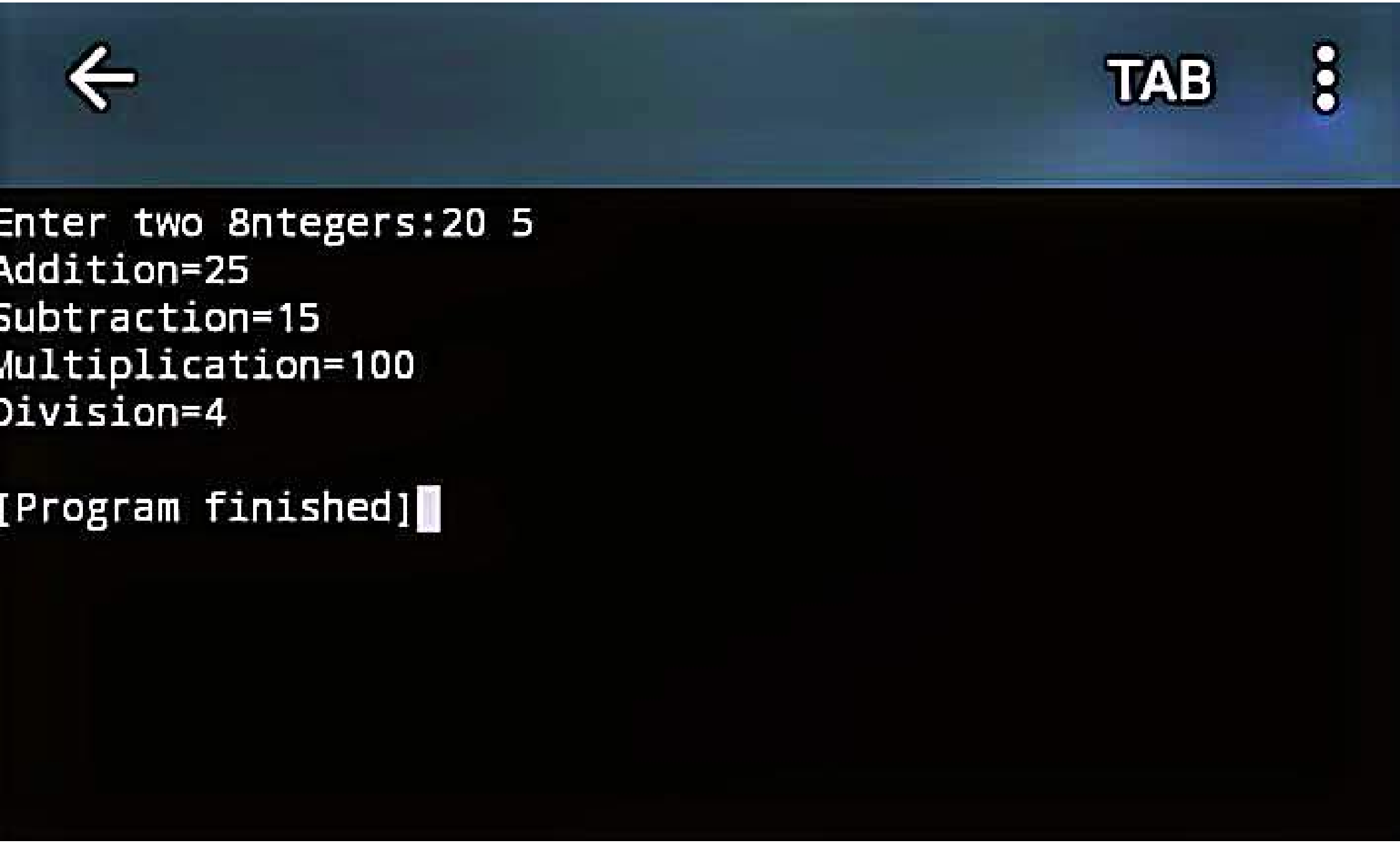
[Process completed - press Enter]
```

CamScanner

new*

```
1  # include <iostream>
2  using namespace std;
3  int main ( )
4  {
5  int num1,num2;
6  cout<<"Enter two 8ntegers:";
7  cin>>num1>>num2;
8  cout<<"Addition="<<num1+num2<<endl;
9  cout<<"Subtraction="<<num1-
   num2<<endl;
10 cout<<"Multiplication="<<num1*
   num2<<endl;
11 cout<<"Division="<<num1/num2<<endl;
12 return 0;
13 }
14
```

CamScanner


TAB
Enter two 8ntegers:20 5
Addition=25
Subtraction=15
Multiplication=100
Division=4
[Program finished]

CamScanner

Output:

```
Enter first number: 10
Enter second number: 5

Sum: 15
Subtraction: 5
Multiplication: 50
Division: 2
```

Task 3:

Write the output of the following code.

```
#include<iostream>
using namespace std;
int main()
{
  int val1;
  int val2;
  int val3;
  cout<<"Please enter your 3 numbers:";
  cin>>val1>>val2>>val3;
  cout<<val1<<"\n"<<val2<<"\n"<<val3<<"\n";
  cout<<val3<<"\n"<<val2<<"\n"<<val1<<"\n";
  return 0;
}
```

Output:

```
10
20
30
30
20
10
```

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Coding C++

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RUN MENU

```cpp
#include <iostream>
using namespace std;

int main()
{
    int val1;
    int val2;
    int val3;

    cout << "Please enter your 3 numbers: ";
    cin >> val1 >> val2 >> val3;

    cout << val1 << "\n" << val2 << "\n" << val3 << "\n";

    cout << val3 << "\n" << val2 << "\n" << val1 << "\n";

    return 0;
}
```

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Compile Result

```
Please enter your 3 numbers: 10 20 30
10
20
30
30
20
10

[Process completed - press Enter]
```

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Task 4:

Write a C++ program to calculate the distance between two points. Note x1, y1, y2 are all double values.

Formula:

$$d = (x_2 - x_1)^2 + (y_2 - y_1)^2$$

Input code:

```
#include<iostream>
using namespace std;
int main(){
    double x1, y1, x2, y2;
    cout<<"Enter x2 y3 x5 y7 :";
    cin>>x1>>y1>>x2>>y2;
    double d = pow(x2-x1,2)+Pow(y2-y1,2);
    cout<<"Distance="<<d;
    return 0;
   }
```

Output:

Enter x2 y3 x5 y7

Distance = 25

Task 5:

i) Calculate the area of a circle (area: PI*r2)

Input code:

```
#include<iostream>
 using namespace std;
 int main(){
     double length, width, area;
     cout<<"Enter length and width: 4 6";
     cin>>length>>width;
     area = length * width;
     cout<<"Area of Rectangle=24"<<area;
```

```
return 0;
}
```

Output:

```
Enter length and width 4 6
Area of Rectangle = 24
```

ii) Calculate the area of a Circle (area = Pi*r2)

```
#include<iostream>
using namespace std;
int main(){
 double r = 5;
 double Pi = 3.1416;
 double area = Pi * r * r;
 cout<< "Enter radius : "<< r << endl;
 cout<< "Area of circle = "<< area;
  return 0;
}
```

Output:

```
Enter radius: 5
Area of Circle = 78.54
```

iii) Calculate the Area of a Triangle (area = base*height*.5)

```
#include<iostream>
 using namespace std;
 int main(){
   double base = 4, height = 5;
   double area = base * height * 0.5;
   cout<< "Enter base and height: "<<base<< " "<<height<<endl;
   cout<< "Area of Triangle = "<<area;
   return 0;

  }
```

Output:

```
Enter base and height: 4 5
Area of Triangle = 10
```

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Coding C++

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RUN MENU

```cpp
#include <iostream>
#include <cmath>
using namespace std;

int main()
{
    double x1, y1, x2, y2;
    double distance;

    cout << "Enter x1: ";
    cin >> x1;

    cout << "Enter y1: ";
    cin >> y1;

    cout << "Enter x2: ";
    cin >> x2;

    cout << "Enter y2: ";
    cin >> y2;

    distance = sqrt((x2 - x1) * (x2 - x1) +
                    (y2 - y1) * (y2 - y1));

    cout << "Distance = " << distance << endl;

    return 0;
}
```

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Compile Result

```
Enter x1: 2 3 5 7
Enter y1: Enter x2: Enter y2: Distance = 5

[Process completed - press Enter]
```

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Coding C++

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RUN MENU

```
    circleArea = 3.14159 * radius * radius;

    // Rectangle
    cout << "Enter length of rectangle: ";
    cin >> length;

    cout << "Enter width of rectangle: ";
    cin >> width;

    rectangleArea = length * width;

    // Triangle
    cout << "Enter base of triangle: ";
    cin >> base;

    cout << "Enter height of triangle: ";
    cin >> height;

    triangleArea = 0.5 * base * height;

    // Results
    cout << "\nArea of Circle = " << circleArea << endl;
    cout << "Area of Rectangle = " << rectangleArea << endl;
    cout << "Area of Triangle = " << triangleArea << endl;

    return 0;
}
```

CamScanner

Compile Result

```
Enter radius of circle: 5 10 4 8 6
Enter length of rectangle: Enter width of r
ectangle: Enter base of triangle: Enter hei
ght of triangle:
Area of Circle = 78.5397
Area of Rectangle = 40
Area of Triangle = 24

[Process completed - press Enter]
```

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Task 6:

Total Purchase A customer in a store is purchasing five items. The prices of the five items are:

Price of item 1 = $ 12.95

Price of item 2 = $ 24.95

Price of item 3 = $ 6.95

Price of item 4 = $ 14.95

Price of item 5 = $ 3.95

Input:

```
#include<iostream>
 using namespace std;
 int main()
     double item = 12.95
     double item = 24.95
     double item = 6.95
      double item = 14.95
      double item = 3.95
     double subtotal = item1+item2+item3+item4+item5;
     double salesTax = subtotal * 0.06; //6% tax
     double total = subtotal + salesTax;
     cout<<"Price of item 1 : $" << endl;
     cout<<" Price of item 2 : $" << endl;
     cout<<" Price of item 3 : $" << endl;
     cout<<"Price of item 4 : $" << endl;
     cout<<"Price of item 5 : $" << endl;
     cout << endl;
     cout << " Subtotal = $" << subtotal << endl;
     cout << " sales Tax (6%) = $" << salestax << endl;
      cout << " Total = $" << endl;
      return 0;
     }
```

Output:

Price of item 1 = $12.95
Price of item 2 = $24.95
Price of item 3 = $6.95
Price of item 4 = $14.95
Price of item 5 = $3.95

Subtotal = $63.75
Sales Tax (6%) = $3.825
Total = $67.575

CamScanner

Coding C++

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RUN MENU

```
    double item1 = 12.95;
    double item2 = 24.95;
    double item3 = 6.95;
    double item4 = 14.95;
    double item5 = 3.95;

    double subtotal;
    double salesTax;
    double total;

    subtotal = item1 + item2 + item3 + item4 + item5;
    salesTax = subtotal * 0.06;
    total = subtotal + salesTax;

    cout << fixed << setprecision(2);

    cout << "Item 1 Price = $" << item1 << endl;
    cout << "Item 2 Price = $" << item2 << endl;
    cout << "Item 3 Price = $" << item3 << endl;
    cout << "Item 4 Price = $" << item4 << endl;
    cout << "Item 5 Price = $" << item5 << endl;

    cout << "Subtotal = $" << subtotal << endl;
    cout << "Sales Tax = $" << salesTax << endl;
    cout << "Total = $" << total << endl;

    return 0;
}
```

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Compile Result

```
Item 1 Price = $12.95
Item 2 Price = $24.95
Item 3 Price = $6.95
Item 4 Price = $14.95
Item 5 Price = $3.95
Subtotal = $63.75
Sales Tax = $3.82
Total = $67.58

[Process completed - press Enter]
```

CamScanner

Coding C++

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RUN MENU

```cpp
#include <iostream>
using namespace std;

int main()
{
    int var1, var2, temp;

    cout << "Enter value of var1: ";
    cin >> var1;

    cout << "Enter value of var2: ";
    cin >> var2;

    cout << "\nBefore swapping:" << endl;
    cout << "var1 = " << var1 << endl;
    cout << "var2 = " << var2 << endl;

    temp = var1;
    var1 = var2;
    var2 = temp;

    cout << "\nAfter swapping:" << endl;
    cout << "var1 = " << var1 << endl;
    cout << "var2 = " << var2 << endl;

    return 0;
}
```

CamScanner

Compile Result

```
Enter value of var1: 10 20
Enter value of var2:
Before swapping:
var1 = 10
var2 = 20

After swapping:
var1 = 20
var2 = 10

[Process completed - press Enter]
```

CamScanner

Coding C++

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RUN MENU

```cpp
#include <iostream>
using namespace std;

int main()
{
    int number;

    cout << "Enter a five-digit number: ";
    cin >> number;

    int digit1 = number / 10000;
    int digit2 = (number / 1000) % 10;
    int digit3 = (number / 100) % 10;
    int digit4 = (number / 10) % 10;
    int digit5 = number % 10;

    cout << digit1 << "    "
         << digit2 << "    "
         << digit3 << "    "
         << digit4 << "    "
         << digit5 << endl;

    return 0;
}
```

CamScanner

Compile Result

```
Enter a five-digit number: 12345
1   2   3   4   5

[Process completed - press Enter]
```

CamScanner

Coding C++

RUN MENU

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```cpp
#include<iostream>
using namespace std;

int main()
{
  char letter;
  cout<<"Enter a lowercase alphabet: ";
  cin>>letter;
  letter=letter - 32;
  cout<<"Uppercase alphabet:"
  <<letter<<endl;
  return 0;
  }
```

CamScanner

Compile Result

```
Enter a lowercase alphabet: b
Uppercase alphabet:B

[Process completed - press Enter]
```

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